

# Keeping Cool at the Third Solar Decathlon

Anyone interested in the latest earth-friendly heating, cooling, appliance, and other home energy technologies should check out the Solar Decathlon, a biennial event sponsored by DOE that invites teams from colleges and universities around the world to design and build cutting-edge green homes. The homes are assembled on the Capitol Mall in Washington, D.C., and teams compete for top honors in ten contests (that's why it's a decathlon). Each home must meet all its operating energy needs, and also produce enough energy to power an electric vehicle, from the sun.

The third Solar Decathlon was held in Washington on October 12–20, 2007. In the closest Solar Decathlon ever, the team from Germany—Technische Universität Darmstadt—took first place; but only by 25 points (out of a total of 1,200 possible points). Similarly, the second place team, University of Maryland, beat the third place team from Santa Clara University by only 20 points. In this contest, the smallest error could make the difference between first and sixth place. In the tale to follow, we learn that embracing and highlighting the right technology—as one of the teams did—pays off. But making a suboptimal technology choice for aesthetic reasons does not.

## Why Not Just Use AC?

Only one team in the 2007 contest—thirteenth-place MIT—used a traditional vapor compression air conditioner like those used in most homes. As the Maryland team put it “the problem with [such] air conditioners is that they are energy hogs.” In what follows I highlight three teams that used innovative, but little-heralded, thermally activated approaches to run dehumidification and cooling technologies in their respective homes. These technologies

were the University of Maryland's Liquid-Desiccant Waterfall and the solar powered absorption chillers used by Santa Clara University and the University of Cincinnati.

These thermally activated technologies are intrinsically efficient—all the energy stays in thermal form. From the Second Law of Thermodynamics, we know that changing from thermal to electric and back to thermal is inherently inefficient. With global warming and global economic growth, cooling demand is rapidly increasing. Between 1999 and 2003, annual sales of air conditioning appliances rose by 31% worldwide, and between 2002 and 2003, such sales more than doubled in Spain (see Figure 1, p. 12).

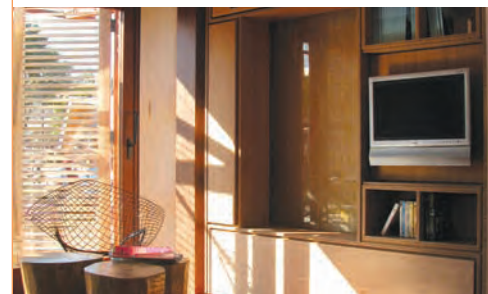
## Desiccant Dehumidification

Dehumidification can consume more than half of the energy used in residential air conditioning in humid areas such as Washington D.C., where the Solar Decathlon takes place. A desiccant absorbs water directly from the air without the need for electrical or thermal energy. But as the desiccant absorbs more and more water, it becomes “full” and stops absorbing. Where solar (or any) heat is available, it makes sense to “regenerate” the desiccant by heating it and letting the water evaporate to the outside air so it can be reused. Desiccants are used for dehumidification in industry, but they are not yet used for residential applications in the United States.

The University of Maryland (UMD) team used calcium chloride (CaCl) as a desiccant. Unlike the designers of commercial desiccant systems, who use a solid desiccant, the UMD team used a liquid desiccant—created by mixing CaCl with water and circulating this liquid with an electric pump through a prominent feature of its house—the indoor waterfall



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(top) The colorful University of Cincinnati house used its 120 extra-large sized Sunda Seido evacuated tubes as a design element—a wall that provides privacy for the deck on the south side of the house in addition to hot water. (bottom) University of Maryland's custom made Liquid Desiccant Waterfall is pictured to the left of the television. The Maryland team took a problem—the need to expose a large surface of the desiccant system indoors—and turned it into an attractive indoor design element.

(see photo, this page). In the UMD house, the solar hot water collectors are evacuated tubes made by Apricus, which were also used by eight other teams in the competition.

Because the desiccant dehumidifies the indoor air, the house's air conditioning could be downsized, and it did not need to run so often, reducing electricity use. UMD also used an energy recovery ventilator to provide fresh air while preserving indoor temperature and humidity levels.

## Absorption Chiller 101

The absorption chiller's thermally activated vapor compression cycle uses heat rather than electricity as an input. Two fluids drive the chiller process. The first is the “refrigerant,” which changes phase to cause the cooling effect. The refrigerant used by the teams is water. The second fluid is the “absorbent”—usually a saline solution. Both Santa Clara University (SCU) and the University of Cincinnati (UC) used a solution of lithium bromide (LiBr) and water as the absorbent.



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The perfect weather during much of the contest contributed to the multiple first places in Hot Water and Energy Balance. The University of Maryland's house, shown above, used a thermally activated dehumidifier to help the university in the cooling-related contests and overall.

At the beginning of the cycle, the refrigerant and absorbent are together in a dilute (weak) solution of LiBr. When the solar hot water reaches a threshold temperature ( $>150^{\circ}\text{F}$ ), the weak solution is pumped into the generator where the solar input provides enough heat to boil the water (refrigerant) out of the weak solution. The water vapor then flows to the condenser, where it is condensed and heat is rejected. The condensed water flows through an expansion device, where the pressure is reduced. The heat flows into the evaporator to evaporate the water (providing the desired cooling effect). The water vapor then returns to the absorber at the desired temperature, somewhere between  $43^{\circ}\text{F}$  and  $50^{\circ}\text{F}$ . The temperature varies a few degrees depending on the ambient conditions, but it is limited on the low end for frost protection. The absorbent is then used to move the cooled water vapor through the system. (If your head hurts from reading this last bit, then just think of the absorber/pump/solution heat exchanger/generator assembly as a replacement for the

electrically powered compressor used in a traditional home AC system.)

At the heart of the UC chilling system is a small absorption chiller made by Rotartica, a manufacturer in Spain. Unlike most absorption chillers, this system has no cooling tower. Instead, the Rotartica has a convection (air) cooled dissipation circuit (that is, a radiator). Rotartica was able to make its small and lightweight absorption chiller by running the absorption cycle in a rotating unit that turns as a result of the static pressure developed by the working fluids. This approach improves the mass and heat transfer enough to make up for its small size.

The solar-thermal hot water comes from 15 Sunda Seido 2-8 evacuated tube arrays along the entire south face of the UC house (see photo, p. 9). The extra-large horizontal Seido 2 tubes were chosen because they can produce extremely (for solar thermal) hot water, which ensures that the absorption chiller runs efficiently. Each tube was rotated individually so that the metal collection plate was  $54^{\circ}$  above horizontal to optimize energy collection at the Washington, D.C. site.

The UC system achieves its design efficiency at a temperature of approximately  $195^{\circ}\text{F}$  but works fairly well down to  $170^{\circ}\text{F}$ . The hot water produced by the evacuated tubes is stored in two custom-built 350-gallon tanks. One of the tanks holds the hottest water directly from the tubes, while another slightly cooler tank receives water from the first as the temperature of the water in the first declines. Water is pumped back through the tubes as needed to maintain a high temperature in the chiller system. The tanks provide heat storage for fast response use at night in the heating season or in the afternoon in the cooling season.

UC's use of two additional small water tanks outside the house is nonstandard. It is a workaround to overcome a mismatch between the

capacity of the absorption chiller and the capacity of the cooling coils in the air-handling unit of the house. The chiller's capacity (1.5 tons) is twice that of the coils (0.75 tons or 2.25 kW), and the chiller's cooling water flow rate is approximately 8 gallons (30 liters) per minute, nearly four times greater than the coil's 2 gallons (8 liters) per minute. In both heating and cooling situations, water is then cycled through a fan coil unit with a water-to-air heat exchanger.

The SCU system includes a prototype chiller based on a commercial chiller made by Yazaki and a flat-plate solar-thermal collector manufactured by Solid. Even though SCU's absorption chiller is the larger of the two, it is driven with hot water from flat-plate collectors, which are considered to be less reliable at delivering the  $150+^{\circ}\text{F}$  minimum needed for chilling (see "Finding a Partnership Between Solar and Energy Efficiency," *HE* 2007 Special Issue: Solar & Efficiency, p. 24). But SCU chose flat panels because their profile and color blended well with the PV arrays (see photo, p. 12). We'll see later that this aesthetic decision led to a non-operational cooling system. SCU also cited the fact that the collectors are packaged in larger sizes ( $6 \times 17 \text{ ft}^2$ ) than conventional flat plate collectors, easing installation and reducing materials use, as a reason to have chosen them.

Like most absorption chillers, SCU's system has a cooling tower (a water-cooled condenser)—with heat rejected from the water to the air by water spray. This is compatible with the forced-air HVAC system used for both heating and cooling. The chilled water is piped in front of the fan unit and cool air is blown over the pipes and into the home. Like UC's system, SCU's system has two water tanks, but for an entirely different reason. SCU's system provides potable hot water as well as water for cooling and heating, and a second tank is needed to separate the potable hot water from the main thermal storage.

### So How Did They Do?

Detailed competition results from the decathlon show how well the three





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The photo above shows Santa Clara University's rooftop solar array under construction. In the foreground and off to the right are SunPower photovoltaic panels. In the back—almost indistinguishable from the solar electrics—are the solar thermal panels made by Solid.

systems did in the events related to thermally activated cooling. It turns out that only Hot Water was a separate contest. Humidity and temperature (heating and cooling), as well as other factors, were considered in the Comfort Zone contest; all the system efficiencies played a role in the Energy Balance contest; and the HVAC system engineering was considered in the overall Engineering contest. Table 1 summarizes the results overall and for the four cooling-system-relevant contests for the three teams whose cooling systems are described in this article.

The perfect weather during much of the contest—plus ever better designed systems—contributed to the multiple first places in Hot Water and Energy Balance. UMD's thermally activated dehumidifier helped the university in the cooling-related contests and overall. But SCU never ran its solar chiller. The solar plate's hot water never got hot enough under the hazy skies to drive the chiller—though they worked well enough to garner SCU one of the many first places in Hot Water. The weather also cooperated enough to prevent the

lack of a chiller from being much of a drawback for SCU.

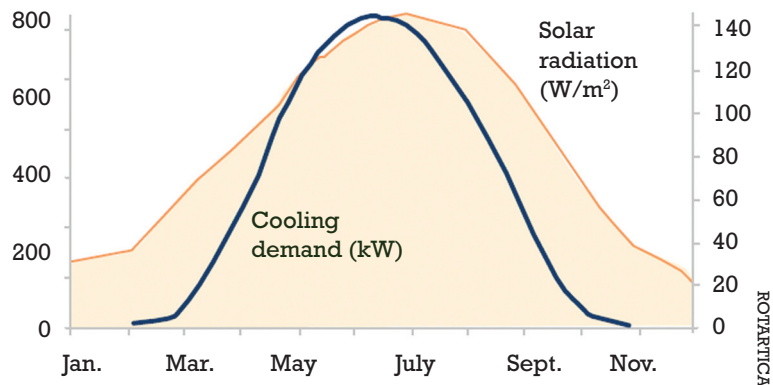
But UC, after tying for first in Energy Balance strayed far out of prescribed temperature ranges at the end of the week, when the weather got hotter. UC had decided, after the first relatively cool day, to save electricity by not running its chiller. Unlike SCU's, UC's solar hot water was more than hot enough to run the chiller, but the chiller's pumps and cooling fans drew far too much electricity—as much as 2.6 kW—that was needed for other contests.

Had the weather stayed cool, this decision might not have hurt UC, but the weather didn't cooperate and UC made one other decision that let aesthetics trump technology. They chose sleek, but non opening windows. Operable windows would have helped cool the house in the second half of the week when temperatures rose, and it was windy. But as DOE Secretary Samuel Bodman said, all the teams were winners, and many lessons were learned about the real-life operation of emerging technologies.

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—Tina Kaarsberg

### Cooling Demand in Spain



**Figure 1.** Between 1999 and 2003, annual sales of air conditioning appliances rose by 31% worldwide, and between 2002 and 2003, such sales more than doubled in Spain

**Table. Competition Results in Five Areas**

| Team | Overall Place | Energy Balance* | Hot Water** | Comfort Zone | Engineering |
|------|---------------|-----------------|-------------|--------------|-------------|
| UMD  | 2             | 1               | 8           | 4            | 6           |
| SCU  | 3             | 1               | 1           | 5            | 10          |
| UC   | 15            | 1               | 9           | 20           | 18          |

\*Seven teams tied for first place in the Energy Balance contest.

\*\*Five teams tied for first place in the Hot Water contest.

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### For more information:

To learn more about the 2007 and the upcoming 2009 Solar Decathlon, go to [www.solardecathlon.org](http://www.solardecathlon.org).

The Product Directory Web page of the 2007 Solar Decathlon provides information on thousands of innovative heating, cooling, appliance, envelope, and other home energy technologies—everything from aerogel floors to zinc roofs. The directory can be found at [www.greenbuildingblocks.com/solar\\_decathlon/product\\_guide.go](http://www.greenbuildingblocks.com/solar_decathlon/product_guide.go).

Additional information about the UMD house can be found at [www.solarteam.org](http://www.solarteam.org).

For more on the UC chilling system, go to <http://solar.uc.edu/solar2007/technology>.

For more on the SCU house and its chiller system, go to [www.scusolar.org](http://www.scusolar.org).